



MINING DEALER BEST PRACTICE

Job Analysis Board

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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**June 2016
1605-1.04-1431**

1.0 Introduction

It was identified that the communication of important information in regards to job scope, job control, planned job duration and cross shift hand-over of component replacement jobs were poor, resulting in extended periods of machine downtime and increased expenses to the MARC (Maintenance And Repair Contract) through inefficiency and scope creep. Progress on the bigger jobs such as component replacements are generally charted on a Gantt Chart for tracking. These are sometimes kept in the Service Office and mechanics do not have visual contact with it readily.

As work is performed on a machine in the workshop, the work includes paperwork such as work order for the repair, work orders for backlog jobs, parts lists, inspection sheets, Service Letters and time cards. These are generally organized as a work pack and placed at any available space on the work bench. They end up being oily and dirty which sometimes get soiled with smudges and become difficult to extract information from the documents.

It is therefore desirable to have a separate board or bench space for the necessary documents to improve job control. WesTrac Equipment had implemented the Job Analysis Board at two MARC sites. It was piloted at Mt Whaleback site with trials made on a whiteboard. The board was then finalised over twelve months and built for each workshop bay. The idea was then adopted at the Boddington site.

As the board was being developed, other items were added. Items such as wheel chocks, step ladders and clean-up utensils were also added to improve labor efficiencies. Relevant safety tags were also added.

2.0 Best Practice Description

This Best Practice describes the use of a Job Analysis Board that has provision for placing different necessary documents for the mechanics' quick reference. It has a small bench for writing and board space for other documents such as the Gantt Chart for the job. All the required clean-up tools are also placed at the back of the board to assist in efficient execution of the job from start to finish.

Figure 1 shows an early version of the job analysis board that was placed on trial. As utilization of the board increased, the layout was modified for improvements. The various personnel involved such a mechanics, leading-hands and supervisors had input to the modifications of the board. Figure 2 shows the final version of the board layout that includes the necessary documents.

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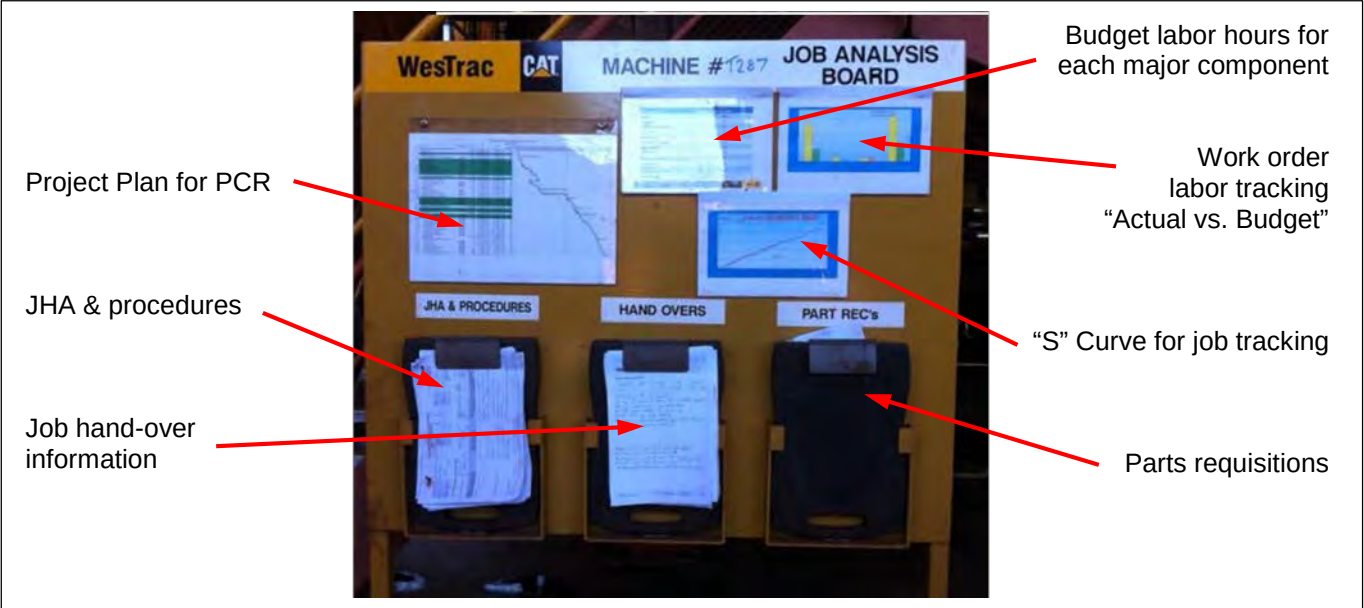


Fig 1: An early version of the Job Analysis Board on trial.

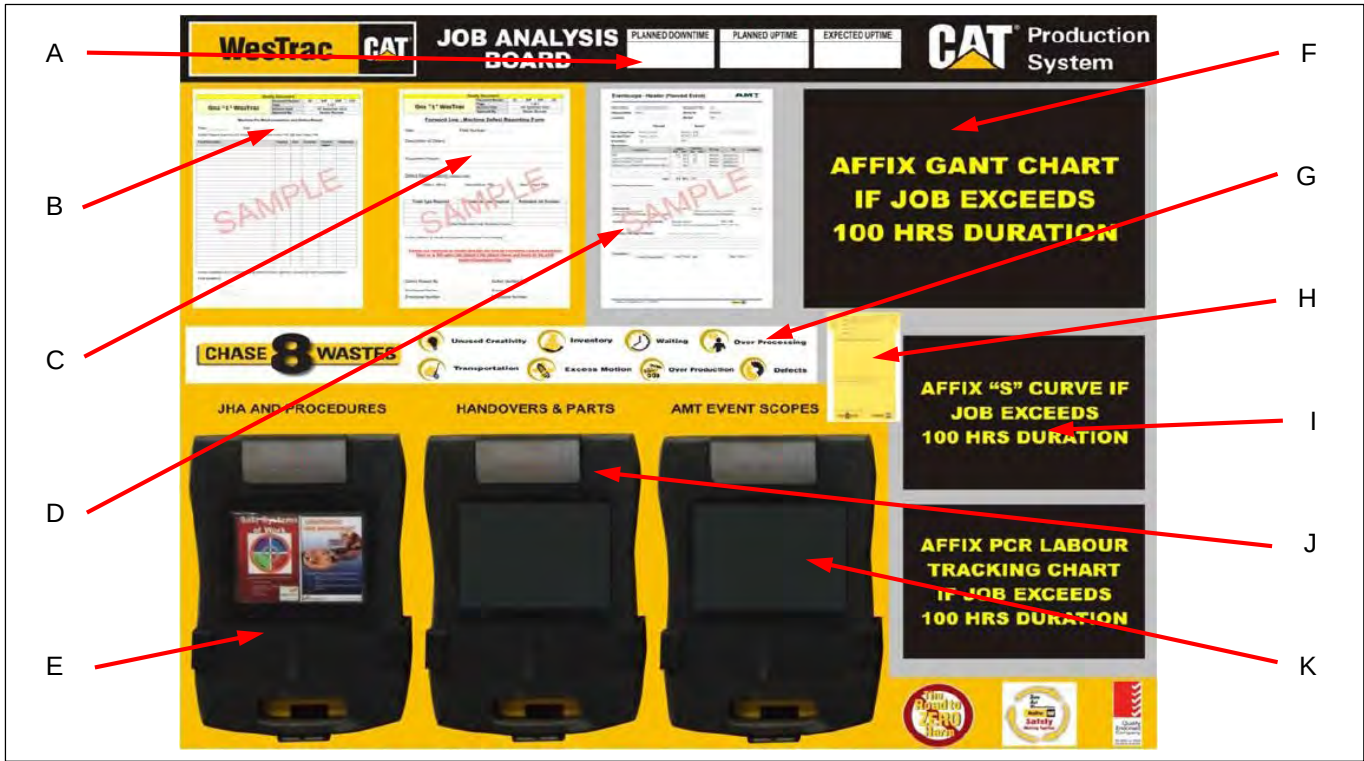


Fig 2: The accepted final version of the Job Analysis Board.

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Reference to figure 2:

- A. **Job Duration Information:** This is a record of the uptime and downtime of planned jobs. The expected uptime will be modified through the duration of the job depending on any delays if the jobs are completed sooner than originally planned. Only the uptime information is entered for breakdown jobs. The 'Planned Downtime' information will just be 'BD' (Break Down).
- B. **Pre-Work Inspection:** Pre-work inspections are conducted as the machine is placed in the workshop. Any defects identified are listed in the form. The Shift Supervisor reviews the identified defects. Only defects related to safety or are expected to cause the machine to have in-service breakdowns will be added to the events scope. All other defects will be placed in the Backlog system for the next opportune time which will generally be the next PM Service.
- C. **Backlog Capture:** The defects identified in the Pre-work inspections to be placed in the Backlog system will be recorded in the Backlog Capture form for each individual defect. These will be converted to jobs for any required parts order and for planning into the next opportune stoppage.
- D. **Events Scope:** The defects identified in the Pre-Work inspections which are not placed in the Backlog system will be converted to additional jobs by the Shift Supervisor. These will be added to the events scope as an updated events scope. The expected uptime will be updated accordingly.
- E. **Safety:** A 'Take 5' is completed to carry out pre-work inspections. After the 'Take 5', all safety documents such as JHA (Job Hazard Analysis) and SWP (Standard Work Procedure) are stored here for reference.
- F. **Gantt Chart:** A Gantt Chart is placed here for jobs that are planned to exceed 100 hours duration. These jobs will generally be major component replacements. The Gantt Chart provides an understanding of the flow of the job and assist in distribution and requirements of other labor such as electricians and welders.
- G. **Chase Waste Reminder:** This is a reminder of the CPS principles of reducing wastes.
- H. **CPS Input Cards:** CPS input cards used for continuous improvements. Any personnel can have input into CPS through the use of these cards.
- I. **Labor Tracker:** For jobs that exceed 100 hours duration, the 'S' Curve chart will be inserted to track actual labor against planned labor.
- J. **Hand-overs & Parts:** All cross shift written hand-over documents are stored here for continuity of the jobs. These will include any SIS print-outs, parts requisition copies and shift change notes.
- K. **Work Scopes:** All work scopes (event scopes) from the work order system is printed and placed here. AMT is the work order system used at these sites.

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Figure 3 shows the front side of a typical unit. This unit shows the set-up at the front just before the unit number was painted on. Each board has a number painted on and each number is of a different color. All the equipment and tools associated to that board are color coded to ensure that they are returned to the right spot.

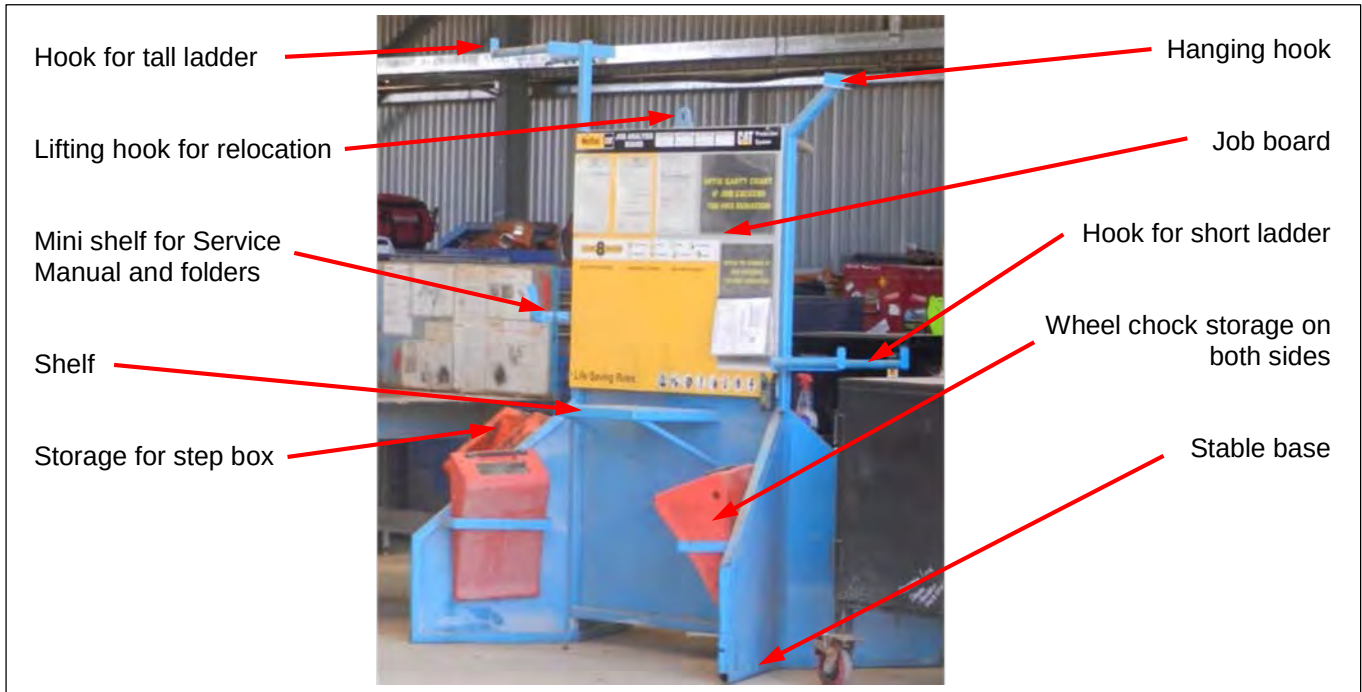


Fig 3: Front side of Job Analysis Board.

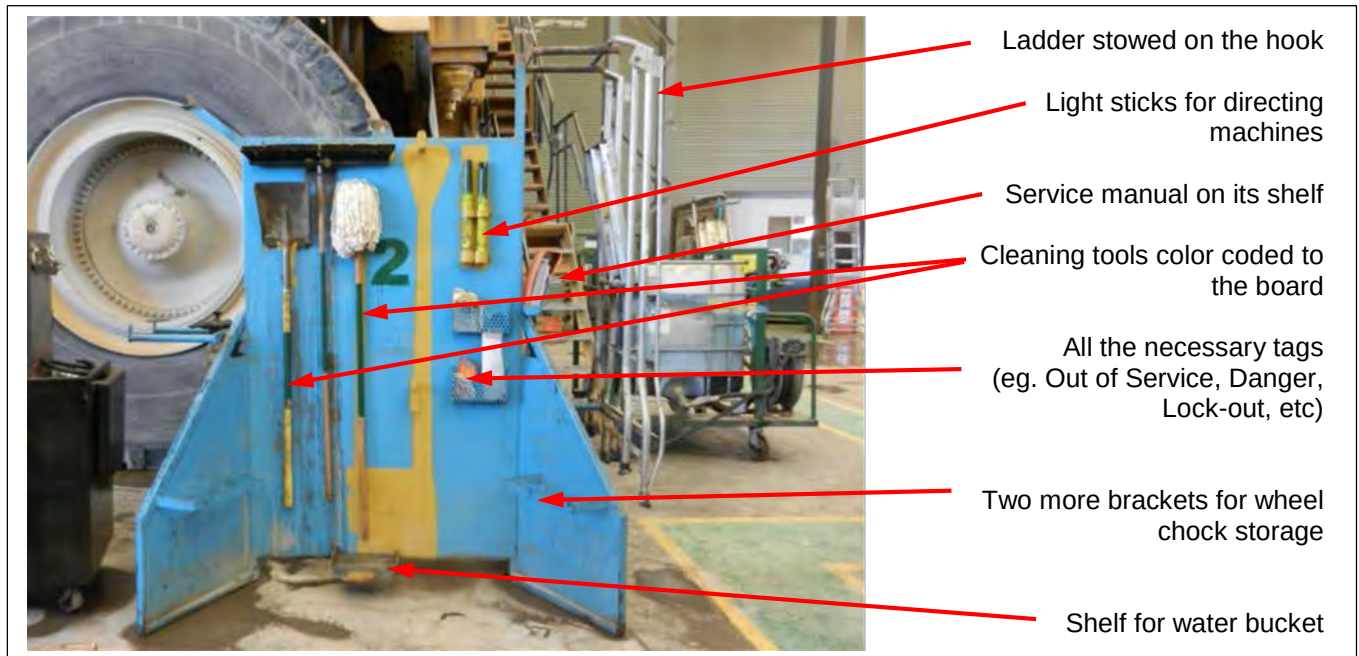


Fig 4: Back side of Job Analysis Board.

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Figure 4 shows the rear side of the unit. All the necessary tags for performing a job are stored here. The clean-up utensils are also stored at the back against its individual 'shadow'. They are color coded to the board. In this case it is a green color for Board Number 2. The number is also painted at the front and sides of the board in green. A quick walk-around at the end of a shift will very quickly identify any missing items.

Figure 5 shows the Job Analysis Boards in use in the workshop. Take note that Board 3 is yellow numbered and Board 5 is red numbered.

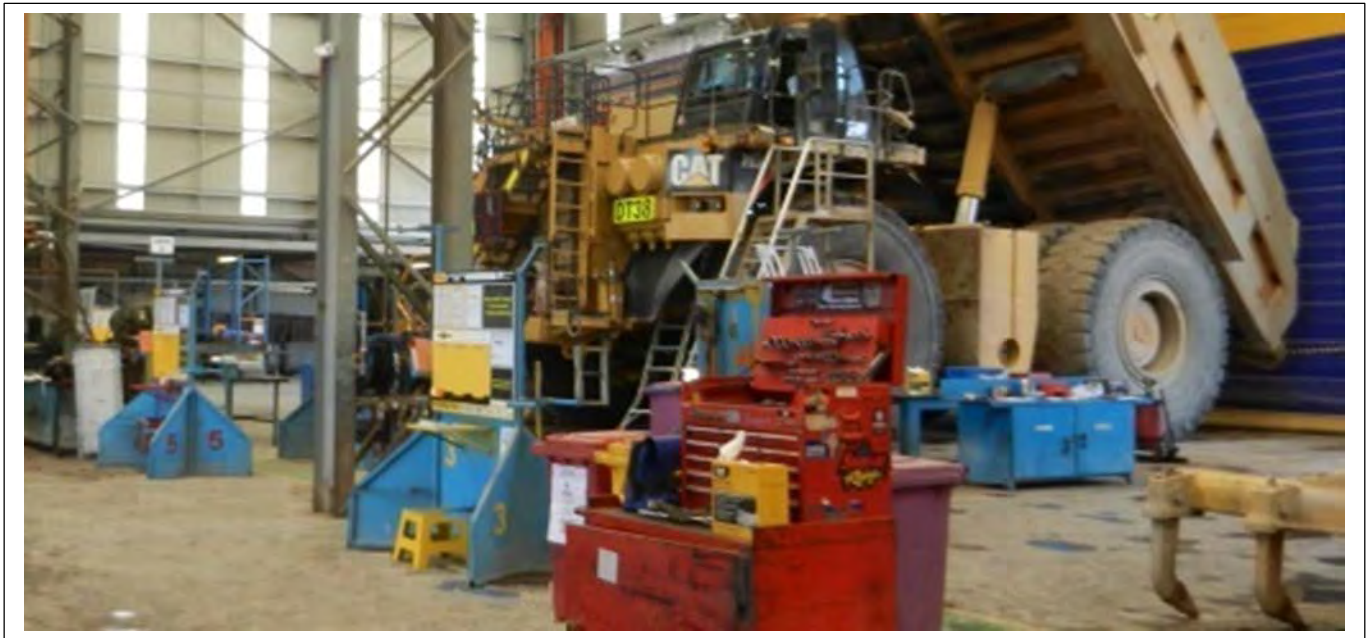


Fig 5: Job Analysis Boards 3 and 5 in use.

3.0 Implementation Steps

Before the manufacturing of the boards are initiated, it must first be decided on what information is required and desired to be available on the board. Below is a general flow process for the implementation steps.

- i. Make a list of the information required to perform the job efficiently. The example in this BP should be a very good start.
- ii. Determine the tags that are required to complete a job from start to finish.
- iii. Determine the utensils required for clean-up during and after a job. Make note of the long items such as brooms and mops.

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- iv. Lay out all the required items at the front and back of the board to determine the size of the board. The height should be such that any personnel of average height will be able to reach all items that require to be attached and detached.
- v. Develop additional required documentation if not available.
- vi. Design the shape of the board for fabrication.
 - a. Suitable height and width dimensions.
 - b. Stability when standing on the workshop floor.
 - c. Ability to be relocated by crane, forklift or personnel. If the board is to have attached castor wheels, they should be heavy duty lockable ones to avoid movements when in use.
- vii. Review the design with workshop personnel. Modify if appropriate.
- viii. Sections of the board should have clear Perspex holders for easy viewing of some documents.
- ix. In general, the whole board is fabricated from steel. If the structure is fabricated from aluminium for weight savings, part of it should have a steel plate for magnetic button holders to keep some documents on the board. These will generally be quick change type of documents for temporary notice.
- x. If desired, part of it can be made of white-board material for writing.
- xi. Select a fabricator for quotation and fabrication.
- xii. Lay out the location for the boards in the workshop.
- xiii. Acquire the equipment for storing with the board. Mark them (color code or otherwise) according to each board.
- xiv. Have it fabricated and delivered with the elected paint scheme.
- xv. Communicate with workshop personnel on implementation of their use.
- xvi. When delivered, set up the board as initially planned.
- xvii. Commence utilizing the boards.
- xviii. Monitor the use of the boards and retrain personnel if required.

4.0 Benefits

Job control is an important part of any work order in a workshop. Lack of job control will introduce wastes in labor utilization and parts usage in many cases. They all potentially contribute to additional downtime. The list below provide a view of some of the benefits of using a job analysis board.

- i. Mechanics have all the necessary tags readily available at one spot near the work bay.
- ii. Mechanics have all the necessary forms and inspection sheets readily availability.

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- iii. The work orders and other relevant documents such as backlog jobs, Service Letters, Special Instructions are in a work pack readily available at the work bay.
- iv. Documents do not get soiled by work benches, which saves time and effort of office personnel when closing the jobs.
- v. The Gantt chart can be easily and readily viewed by mechanics and workshop personnel for information on progress of the job, especially for larger jobs such as component replacements.
- vi. Mechanics save time with readily available wheel chocks at the board and not having to get them from their stowage area.
- vii. Other commonly used access equipment such as ladders and step box are also readily available.
- viii. When the job is completed, valuable time is saved when all the equipment used can be easily placed back on the board.
- ix. In cases of any spills and when the machine had been moved out after a job is completed, the general clean-up equipment is readily available.
- x. Job control and work flow are improved.
- xi. There is a higher visibility of the jobs which promotes compliance to plan.
- xii. Communication at shift hand-overs are improved.
- xiii. The readily available documents and tags invokes higher safety awareness.
- xiv. Closer adherence to procedures is promoted.
- xv. Removes waste and challenges employees to use the CI system.
- xvi. Downtime is reduced through the all the above.

5.0 Resources Required

In almost all mine sites, most or all the required documents are already existing. The Planner needs to review the required documents list for the board and develop any desired ones that are not yet available. He/she can then develop the shape and size of the board to accommodate all the required documents and equipment.

The Project Manager/Maintenance Superintendent will set up an awareness program for the use of the board. This can be communicated to all workshop personnel through shift change meetings and any other relevant meetings. A supplier will have to be appointed.

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6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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